

Ali Hashemi | Ph.D. | Machine Learning Researcher

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- Skilled in GenAI & LLMs, Representation Learning (Diffusion), and Efficient Foundation Models (Distillation, PEFT, Sparsity)
- NeurIPS-published, Strong in theory & software engineering (Optimization, PyTorch, AWS) • Berlin-based, open to relocate

Summary

Proven expertise in building foundation models and LLMs from scratch and optimizing them for efficiency and inference through representation learning (distillation, sparse autoencoders), parameter-efficient fine-tuning, and sparsity methods. Strong theoretical and practical foundation in generative modeling, particularly diffusion models, with expertise in Bayesian statistics, graphical models, and nonlinear optimization, resulting in 10+ top-tier publications. Experienced software engineer proficient in Python, PyTorch, Git, Docker, and distributed multi-GPU training with scalable AI deployment on AWS, including parallel computing across clusters. Specialized in signal and image processing using sparse coding and dictionary learning to achieve efficient, interpretable solutions with applications in healthcare. Successfully led projects and collaborated with interdisciplinary, cross-functional teams to advance multimodal learning and generative AI in biomedical and vision-language domains.

Experiences

- Oct 2022 – Nov 2025
- Machine Learning Research Scientist**, *BIFOLD*, Berlin, Germany.
 - Developed an interpretable **subspace-based distillation** method to build compressed foundation models from scratch, using 600M+ parameter teacher models, achieving 80% size reduction with only 1% accuracy loss.
 - Built and optimized histopathology foundation models using our developed distillation approach on 100M image patches from gigapixel-scale slides, achieving $\uparrow 60\%$ speed, $\downarrow 40\%$ memory, with minimal (1–2%) accuracy loss.
 - Built a **multimodal RAG** system for clinical reporting using contrastive learning on histopathology images and reports, integrating CLIP, Hugging Face Transformers, PEFT adapters, and domain-specific models (e.g., BioGPT), with vLLM/TensorRT/ONNX for low-latency serving—improving reporting efficiency by 15%.

Foundation Models Efficient Inference Distributed Training SAE Distillation PEFT Sparsity Transformers
- Oct 2022 – Nov 2025
- Post-Doctoral Research Fellow**, *Machine Learning Group, Technische Universität Berlin*, Germany.
 - Designed conditional **diffusion models** for biomedical inverse problems and accelerated inference with **distilled consistency models**, enabling near single-step sampling without quality degradation.
 - Developed a hybrid physics-informed neural network that learns structured spatiotemporal representations from brain signals, improving 3D localization accuracy by 12% across 500K+ time-series samples.
 - Organized and taught courses with > 200 participants over 2 years. Supervised 5+ M.Sc. and Ph.D. students.
- Jun 2021 – Sep 2022
- Applied Scientist**, *Quality in Artificial Intelligence (QAI) Labs*, Berlin, Germany.
 - Developed a Riemannian geometry-based optimization algorithm, improving neuroimaging localization accuracy by 9% compared to benchmarks; presented at **NeurIPS** and published in **TMI** (Impact Factor > 10).
 - Directed the development of an open-source brain inverse solutions **software package**, implementing GitHub Actions CI/CD pipelines, Dockerization, unit testing, and collaborative Git workflows, enhancing noise tolerance and diagnostics; widely adopted as a benchmark in the neuroimaging community and available on **GitHub**.
- Feb 2019 – May 2021
- Applied Scientist**, *Brain and Data Science Lab | Charité*, Berlin, Germany.
 - Built hierarchical Bayesian models with sparse priors for **latent representation learning**, enhancing robustness to noise by 20% in clinical brain signal data.
 - Reduced pipeline training time by 30% by engineering reproducible, distributed training workflows using PyTorch, SLURM, Hydra-Zen, and Submitit, with W&B monitoring for performance tracking in clinical studies.
- Feb 2016 – Sep 2022
- Doctoral Research Fellow**, *Machine Learning Group, Technische Universität Berlin*, Germany.
 - Developed **large-scale optimization algorithms** for high-dim settings using sparse Bayesian statistics and ML.
 - Published **over 10 publications** in prestigious journals and conferences.
 - Presented research at esteemed international conferences and workshops including **NeurIPS** and **Med-NeurIPS**.
- Mar 2015 – Jan 2016
- Applied Scientist**, *Singapore University of Technology and Design*, Singapore.
 - Designed SOTA thermal field monitoring techniques by integrating PDEs into **compressed sensing** algorithm, achieved 10 dB improvement in reconstruction accuracy with 90% less computational complexity.
 - Developed a dictionary learning approach for brain classification, achieving 80% compression with no loss in accuracy.

Education

- 2016 – 2022
- Ph.D.–Technische Universität Berlin**, *Computer Science and Machine Learning*, Germany.
Thesis title: “Advances in Hierarchical Bayesian Learning with Applications to Neuroimaging”
Supervisors: Prof. Dr. Klaus-Robert Müller and Prof. Dr. Stefan Haufe.
Grade: Summa cum laude (**with Highest Distinction**).

Technical & Transferable Skills

- Programming
- Python (strongest), C/C++, Rust, R, MATLAB, SQL, $\text{\LaTeX}$
- Libraries & Frameworks
- NumPy, Pandas, SciPy, Scikit-learn, OpenCV, NLTK, Hugging face (Transformers, Adapters, Diffusers), PyTorch (Lightning), TensorFlow, Jax, SLURM, Hydra, W&B, Weaviate
- MLOps & LLM
- Linux, Bash, Git, CI/CD, Docker, Kubernetes, MLFlow, AWS (EC2, S3, SageMaker, Lambda, Bedrock), GCP (Vertex AI), Azure (SDK), LlamaIndex, LangChain, Milvus, Neo4j, TGI, NIM, ONNX, vLLM, TensorRT
- Soft Skills
- Project Management, Technical Writing, Public Speaking, Training & Mentoring, Event Organization

Honors and Awards

- 2018 **Awarded a 4-year scholarship**, *Machine Learning Group*, Technische Universität Berlin, Germany.
- 2016 **Awarded a 3-year full fellowship**, *Berlin International Graduate School in Model and Simulation-based Research (BIMoS)*, Technische Universität Berlin, Germany, [Link].
- 2015 **Awarded a 1-year research scholarship**, *Singapore University of Technology and Design*, Singapore.
- 2011 **Ranked 1st**, among 200+ electrical and computer engineering undergraduate students.
- 2010 **Ranked 10th/30,000**, *National Electrical Engineering Olympiad for Undergraduate Students*.

Publications

- Selected **“Efficient hierarchical Bayesian inference for spatio-temporal regression models in neuroimaging”**, with Y. Gao, C. Cai, S. Ghosh, K.-R. Müller, S. Nagarajan, & S. Haufe, *Advances in Neural Information Processing Systems (NeurIPS)*, 34, 24855–24870, 2021, [DOI | arXiv].
- “Joint hierarchical Bayesian learning of full-structure noise for brain source imaging”**, with C. Cai, K.-R. Müller, S. Nagarajan, & S. Haufe, 34th Conference on Neural Information Processing Systems (NeurIPS), Medical Imaging meets NeurIPS (**Med-NeurIPS**) Workshop, 2020, [Link].
- “Unification of sparse Bayesian learning algorithms for electromagnetic brain imaging with the majorization minimization framework”**, with C. Cai, C. Kutyniok, K.-R. Müller, S. Nagarajan, & S. Haufe, *NeuroImage*, 239, 118309, 2021, [DOI | bioRxiv].
- “Joint learning of full-structure noise in hierarchical Bayesian regression models”**, with C. Cai, Y. Gao, S. Ghosh, K.-R. Müller, S. Nagarajan, & S. Haufe, *NeuroImage*, 239, 118309, 2023, [DOI | bioRxiv].
- Preprints **“Turning students into domain experts via task-relevant knowledge distillation”**, with P. Chormai, K.-R. Müller, & G. Montavon, Preparing to submit to **IEEE PAMI**, 2025. (Impact Factor > 20)
- “Continual learning that resists to overfitting and spurious correlations through low-rank and sparse adaptation”**, with P. Chormai, K.-R. Müller, & G. Montavon, Preparing to submit to **ICML**, 2026.
- “Enhancing brain source reconstruction through physics-informed 3D neural networks”**, with M. Morik, K.-R. Müller, S. Haufe, & S. Nakajima, Submitted to **IEEE TMI**, 2025.
- “How forgiving are M/EEG inverse solutions to noise level misspecification? an excursion into the BSI zoo”**, with A. Negi, S. Haufe, & A. Gramfort, Submitted to **IEEE TBME**, 2025.
- Books & Monographs **“Advances in hierarchical Bayesian learning with applications to neuroimaging”**, Ph.D. Thesis, Technische Universität Berlin, 2022, [DOI].
- “Compressed sensing: From theory to praxis”**, with A. Flinth, & G. Kutyniok, In: C. Chen (Ed.), *Compressive sensing of earth observations*, CRC Press, pp. 1–32, 2017, [DOI].
- Journals **“Bayesian adaptive beamformer for robust electromagnetic brain imaging of correlated sources in high spatial resolution”**, with C. Cai, Y. Long, S. Ghosh, Y. Gao, M. Diwakar, S. Haufe, K. Sekihara, W. Wu, & S. Nagarajan, *IEEE Transactions on Medical Imaging*, 42, 2502–2512, 2023, [DOI].
- “Bayesian algorithms for joint estimation of brain activity and noise in electromagnetic imaging”**, with C. Cai, H. Kang, D. Chen, Y. Gao, M. Diwakar, S. Haufe, K. Sekihara, W. Wu, & S. Nagarajan, *IEEE Transactions on Medical Imaging*, 42(3), 762–773, 2023, [DOI].
- “Empirical Bayesian localization of event-related time-frequency neural activity dynamics”**, with C. Cai, L. Hinkley, Y. Gao, S. Haufe, K. Sekihara, & S. Nagarajan, *NeuroImage*, 258, 119369, 2022, [DOI].
- “Robust estimation of noise for electromagnetic brain imaging with the Champagne algorithm”**, with C. Cai, M. Diwakar, S. Haufe, K. Sekihara, & S. Nagarajan, *NeuroImage*, 225, 117411, 2021, [DOI].

Selected Talks

- Dec 2021 **35th Conference on Neural Information Processing Systems (NeurIPS 2021)**, *Virtual due to Covid*.
Talk on “Efficient Hierarchical Bayesian Inference for Spatio-temporal Regression Models in Neuroimaging.”
- Dec 2020 **34th Conference on Neural Information Processing Systems (NeurIPS 2020)**, *Virtual due to Covid*.
Talk on “Joint Hierarchical Bayesian Learning of Full-structure Noise for Brain Source Imaging.”
- Aug 2022 **22nd International Conference on Biomagnetism (BIOMAG)**, *University of Birmingham, UK*.
Invited Talk on “Hierarchical Bayesian Learning for Noise Estimation in Electromagnetic Brain Source Imaging.”
- Jul 2022 **Artificial Intelligence (AI) for Signal and Image Processing (AI4SIP)**, *Paris Saclay University, France*.
Invited Talk on “Brain Source Imaging: A Benchmarking Approach of Sparse Inverse Problems.”
- Dec 2021 **Metrology for AI in Medicine Lecture Series**, *PTB, Germany*.
Invited Talk on “Numerical Optimization for Machine Learning.”

Languages

English (Fluent), German (Advanced | B2), Persian (Native), Arabic (Basic |A1), Turkish (Basic |A1)